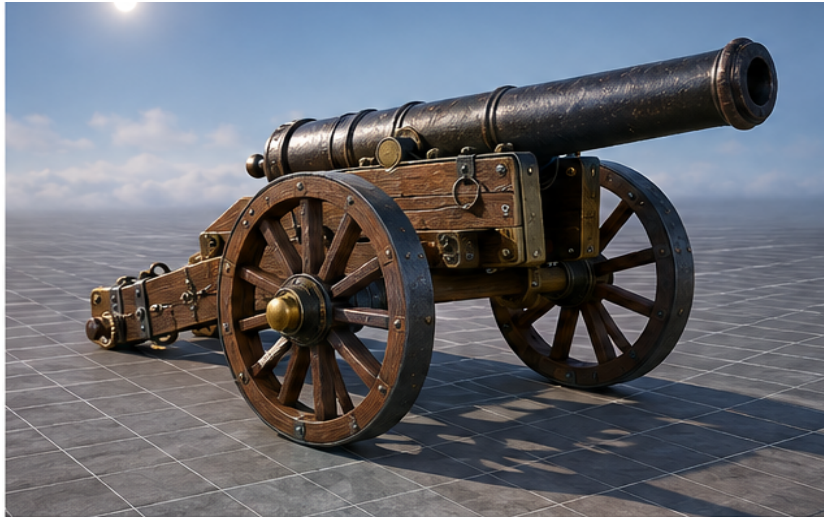


A Medieval Field Cannon in OpenGL

Hierarchical Transformations, Rolling Constraints and the Phong Reflection Model

Name: Sheikh Md. Galib Mahim **Roll:** 2107020



1. What the Project is About

This project presents a small interactive medieval field cannon scene developed in OpenGL to demonstrate fundamental computer graphics concepts through a physically meaningful object. The cannon is modeled as a hierarchy of connected parts: the carriage acts as the root, while the wheels, barrel mount, trunnions and barrel inherit its transformations. The project focuses on translation, rotation, hierarchical transformations, perspective projection, lighting, materials and animation.

The cannon can be driven across the ground, with the wheels rotating according to the rolling without slipping constraint. The barrel can be elevated around its trunnion pivot and recoils along its own rotated axis when fired. A projectile is launched from the muzzle and follows a gravity-based trajectory. Per-fragment Phong shading is used to show contrasting appearances for wooden and metallic components, while fog, a reference grid and an orbiting light enhance the scene.

Core graphics concepts: translation, wheel rotation, barrel elevation, hierarchical and composite transformations, perspective camera, Phong lighting, material properties, continuous animation and user interaction.

2. What Objects Will Be There

- **Cannon barrel:** Cast-iron barrel with adjustable elevation and recoil.
- **Wooden carriage:** Main supporting structure and hierarchical transformation root.
- **Two wheels:** Spoked wheels that rotate consistently with the cannon's movement.
- **Brass fittings:** Metallic fittings used to provide material contrast.
- **Cannonball:** Projectile launched from the muzzle with gravity-based motion.
- **Ground plane and reference grid:** Provides the scene floor and spatial reference.
- **Sky/background and fog:** Creates depth and atmosphere in the scene.
- **Light source:** Orbiting point light used for illumination.
- **Perspective camera:** Orbit camera for viewing and navigating the scene.

3. Extra Feature

- Interactive controls for driving, barrel elevation, camera movement and scene settings.
- Recoil animation using a damped harmonic oscillator model.
- Projectile simulation with gravity and bouncing using a restitution factor.
- Switchable shading and rendering options, including Phong/Blinn comparison and wireframe mode.
- Lighting, material and fog controls to demonstrate different visual effects.

Expected Outcome

The final result will be an interactive OpenGL cannon scene that visually demonstrates how hierarchical transformations, physically constrained motion, projectile animation and the Phong reflection model work together in a complete computer graphics application.